

INTRODUCTION PROJECT MANAGEMENT

Project Management (PMAE/PM)

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Learning objectives

- Be able to recognize projects
- Be able to evaluate formulations for goals
- Be able to create optimized formulations



WHAT IS A PROJECT?

Task in everyday work

- You work as a developer for a software company, TinySoft, which develops software created.

What are their tasks?

- What characterizes your typical workday?
- What is your work environment like?



Werden Sie
aktiv!

Task in everyday work – New request

- You work for a software company, WinzigWeich, which develops C++ programs. A client now wants an Android app to read vehicle data from a car. to be able to.
- What is the difference between this and your normal work?
- What information do you need to complete this task? can?
- Is the assignment a project? If so, why?



Werden Sie
aktiv!

Project vs. Routine

Routine work

- old / familiar destinations
- well-established team
- known processes
- permanent staff
- Festes Budget
- Low risk

project

- new goals
- Newly defined responsibilities within the team
- new / planned processes
- Employees need to be reassigned
- Cost planning and control
- Uncertain, higher risk

Project definition

- A project is a goal-oriented, time-limited undertaking to create a novel product or service
- Characteristics:
 - Clarity of goal
 - Uniqueness and novelty
 - Difficulty of the task
 - Process-oriented nature with many work steps
 - Scheduling (Limited time)
 - Team building (participation of several people)
 - Resource limitations

Project – Yes or No?

- New aircraft
develop and
build



Project – Yes or No?

- Construction of the Elbe-
philharmonic



Image: Pixabay

Project – Yes or No?

- Conducting a
- Flight from Nuremberg
- to Munich



Project – Yes or No?

- Manufacturing an automobile



Image: Audi

Project – Yes or No?

- Setting up a workshop
for manufacturing a new
vehicle



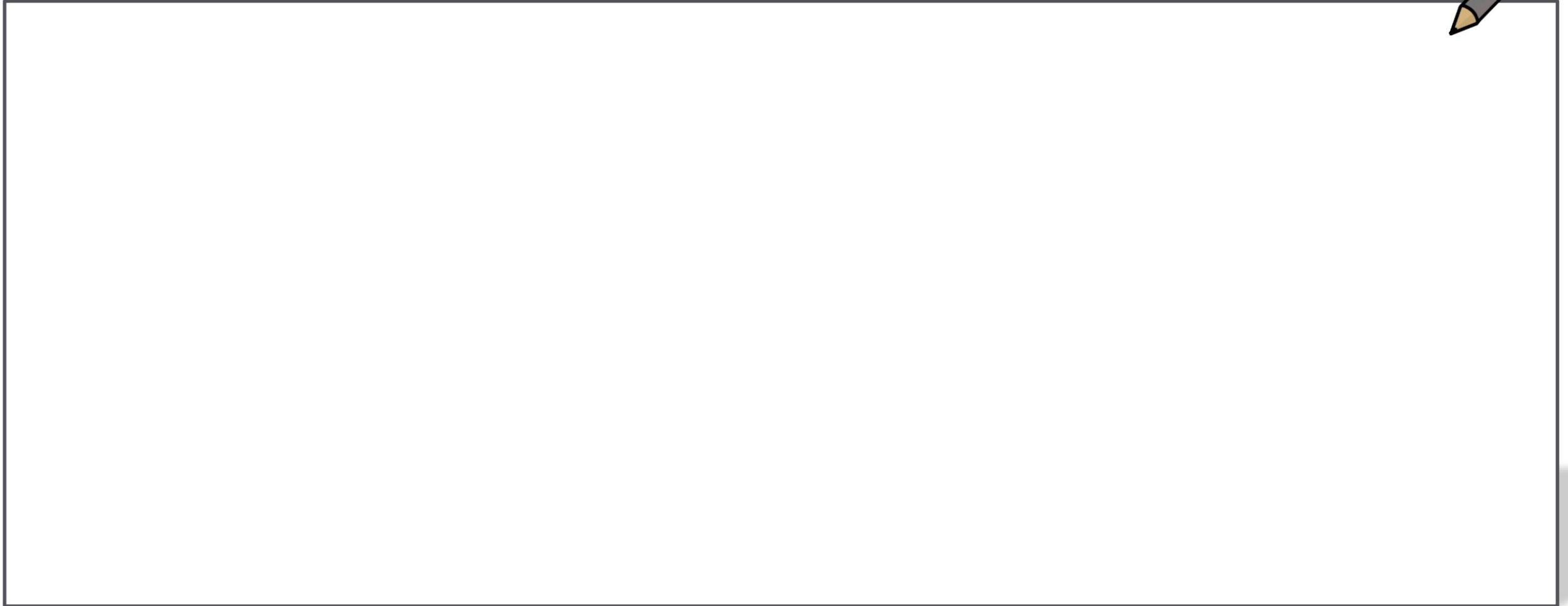
Image: Audi

Project types (examples)

- **Research project** : A project with a high degree of innovation, abstract goals, and high uncertainty
(Exploration of new things)
- **Development project** : A project with a high degree of novelty and concrete objectives, but lower
Uncertainty (compared to research)
- **Project planning project** : Project to create a new system / software from existing modules (few /
(low level of novelty) with a clear scope and goal
- **Organizational project**: Changing operational processes or creating new processes
- **Investment project**: Construction of buildings/roads with a high cost budget due to necessary resources

WHY PROJECT MANAGEMENT?

Reasons for failure: What measures could prevent failure?

A large, empty rectangular box with a thin black border, intended for writing or drawing.

What is project management?

- Project management ..
 - Defines project boundaries and project goals
 - Periodic review of targets
 - process-oriented structuring of projects
 - Design of project organization and project culture
 - Development of project culture
 - Development of the project in the context of the client
- Project management ensures project success by taking into account:
Corporate strategy and competitive situation secure

Definition of Project Management (PM)

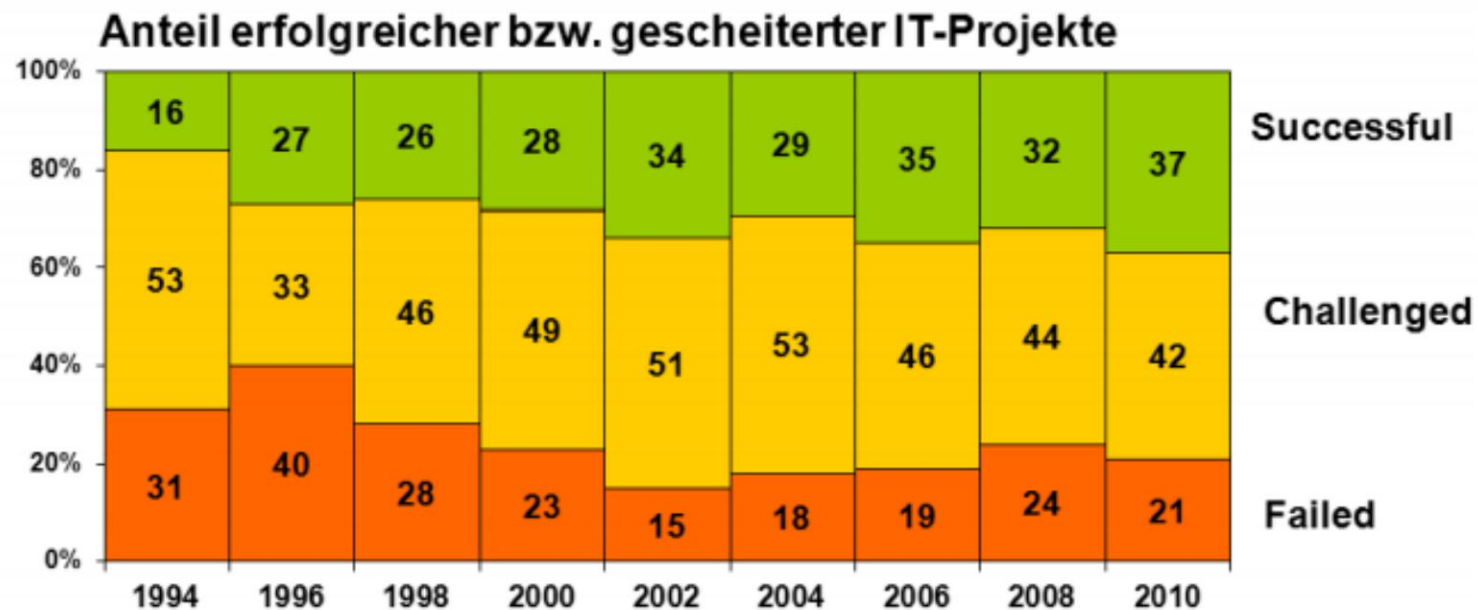
- Project management includes the initiation, control, **planning**, monitoring, and completion of a project.
- PM is the **application of knowledge, skills, tools** and techniques, including project activities, to achieve
To meet project requirements
- Project management is the **guidance of project stakeholders** to ensure the safe achievement of project goals.

Why project management?

Standish Group Der Chaos Report

Institut für kreatives
Management und Training

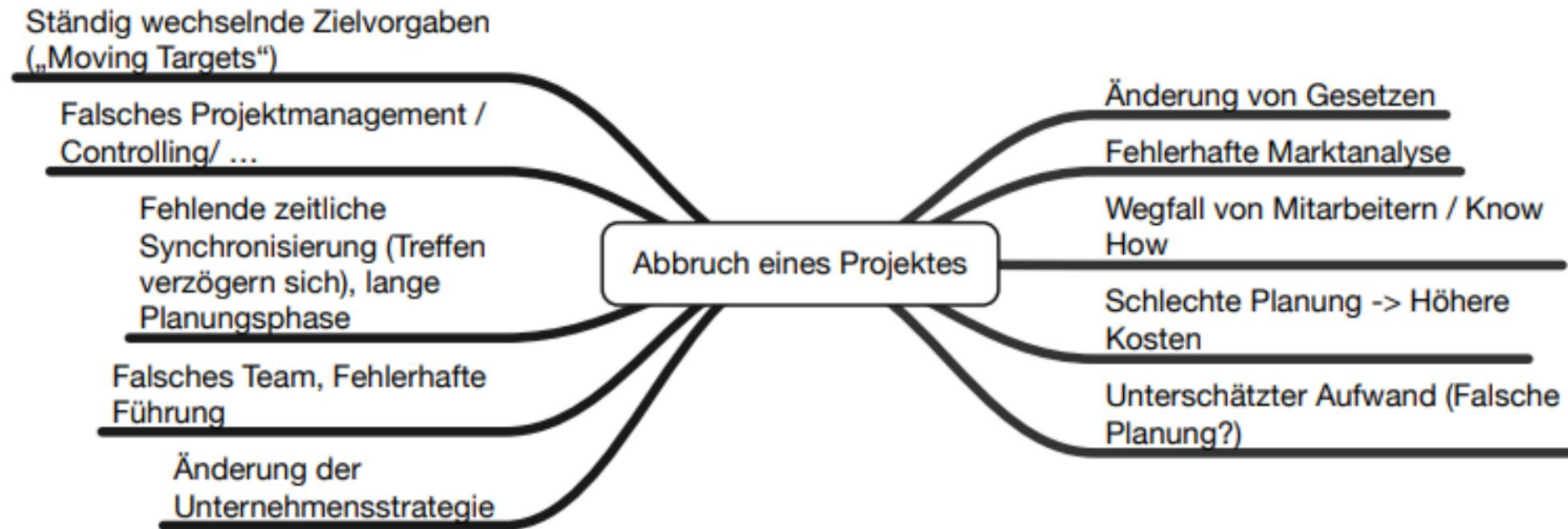
IKMT



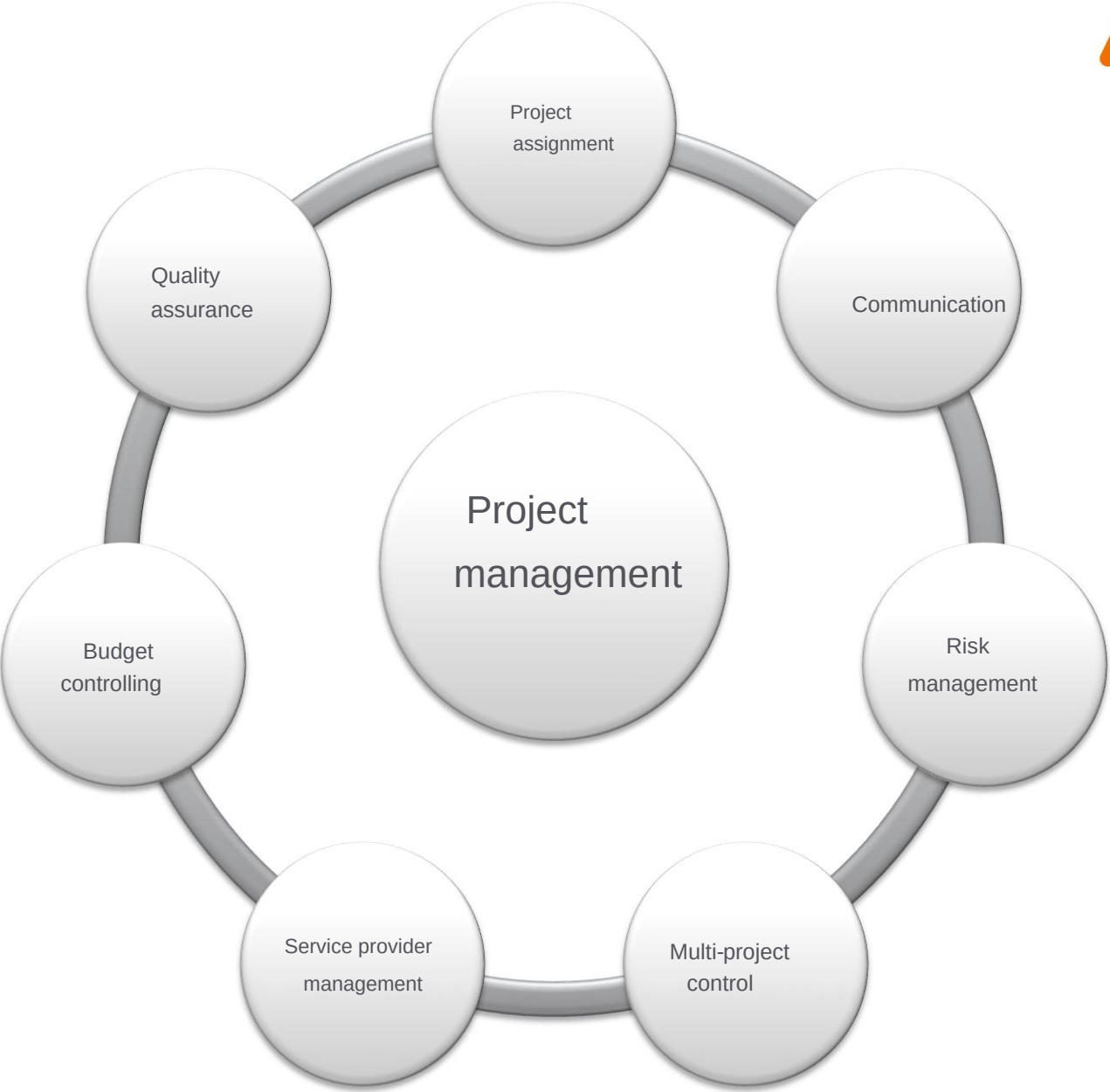
Quelle: Standish Group - Chaos Report

Reasons for failure

- 1/3 of IT projects are cancelled
- Almost half of IT projects are delayed.



Components



PROJECT GOALS

Project goals

- Goals...
 - define a question of meaning (Why is something being done?) – motivation
 - define priorities + targeted action (in order!)
 - Enabling delegation (Who can do something (better)?)
 - Directional guidance for the current actual state and target state (goal!)
 - Used for performance monitoring
- Problem: A goal must be presented **in a concrete way** !
- Depending on how the goal is described, this has a positive or negative effect.

Good or bad project goals?

- a. Make the website look nice by Friday
- b. I will study for 2 hours every day for the exams coming up at the end of the month.
- c. We should improve project processes.
- d. I want to earn 5000 euros per month by the end of the year.

Defining project goals - assistance



Better:

ATTRACTIVE

(Image: Lochbronner.com)

Which of the following goals are SMART?

-
- For each goal, note on your smart device/notepad which of the requirements apply to the
The objective has been achieved.
 - You have 15-60 seconds for each task
 - Write the optimally formulated goal on the respective slide.

Is the following goal formulated in the SMART way?

-
- I will study for 2 hours every day for the exams coming up at the end of the month.



The image shows the word 'SMART' in a large, bold, sans-serif font. Each letter is white and is contained within a light blue square. The squares are arranged horizontally and are slightly separated from each other. The background is a very light gray.

Is the following goal formulated in the SMART way?

-
- We should improve project processes.



The image shows the word 'SMART' in a large, bold, sans-serif font. Each letter is white and is contained within a light blue square. The squares are arranged horizontally and are slightly separated from each other. The entire graphic is set against a light gray background.

Is the following goal formulated in the SMART way?

-
- I want to earn 5000 euros a month.



The image shows the word 'SMART' in a large, bold, sans-serif font. Each letter is white and is contained within a light blue square. The squares are arranged horizontally and are slightly separated from each other. The background is a very light gray.

Is the following goal formulated in the SMART way?

-
- In order to reduce our inventory, motorcycle production should be increased from 800 to 850 units.
will be increased within 4 weeks.

S M A R T

Is the following goal formulated in the SMART way?

-
- We need to increase customer satisfaction



The image shows the acronym 'SMART' in a large, bold, sans-serif font. Each letter is white and is contained within a light blue square. The squares are arranged horizontally and are slightly separated from each other. The entire graphic is set against a light gray background.

Is the following goal formulated in the SMART way?

-
- To reduce stress, we should create a plan by Friday to make the project more effective
edit



The image shows the word 'SMART' in a large, bold, sans-serif font. Each letter is white and is contained within a blue square. The squares are arranged horizontally and are slightly separated from each other. The background is a light gray.

Project objectives – definition according to DIN 69901

- DIN 69901 Project objectives:
 - ... as **a verifiable result** and given boundary conditions of the overall task of a project
 - ... **the entirety of individual goals** that will be achieved through the project.
- Goal definition: **Quantitative and qualitative definition** of the project content and the implementation conditions to be met, e.g., costs and duration, in target characteristics with usually different target weights.

Why is a good goal important?

- **Planning:** When the goal has been specifically defined, planning is much easier.
- **Clarification:** Goals provide an overview of what has been done and what still needs to be done.
- **Regulation of obligations:** Defining one's own and others' tasks is only possible through objectives.
possible

Why is a good goal important?

- **Orientation:** A project is in flux, i.e., ongoing changes; goals allow for an assessment of whether work should be carried out.
Is it effective or not?
- **Adaptation:** Significant changes in framework conditions (and potentially unattainable goals) can lead to
Adjustments lead to
- **Motivation:** Goals point the way and allow you to see the BIG PICTURE - the reason for the activity becomes visible.
• Increased motivation



Why is a good goal important?

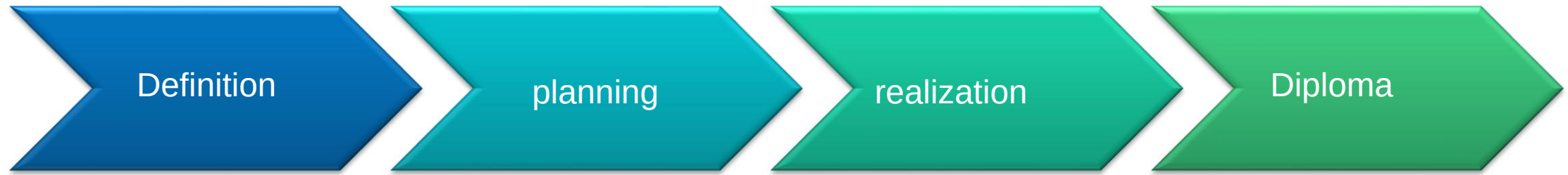
- **Communication:** Project goals make it possible to talk about and discuss parts of the project.
- **Evaluation:** Clear targets are linked to success criteria that can be evaluated.

Prioritizing goals

- **Mandatory objectives:** contractually agreed-upon objectives with the highest priority. These objectives are the reasons for a project and define the success of the project (minimum project scope)
- **Target goals:** Goals should be achieved if possible. Desired goals category, but may be modified if necessary.
Changes will be made after consultation.
- **Optional goals:** Improvement of the overall result - Optional and have no impact on the success of the Result
- Priority: Must ÿ Should ÿ Can
- These categories should be assigned when creating goals.

PROJECT LIFE PHASES

Project life cycle phases: Definition phase



- a) Acquisition
- b) Zieldefinition
- c) Meilensteinplan
- d) Coarse
Cost plan
- e) Feasibility

Project life cycle phases: Planning phase



- a) Structure Plan
- b) Work packages
- c) Cost plan
- d) Resource need
- e) GANTT-Chart

Project life cycle phases: Implementation phase



- a) Progress monitoring
- b) Cost control
- c) Milestone-
Meetings
- d) Weekly Meetings

Project life cycle phases: Completion



- a) handover
Results
- b) Final meeting
- c) Final invoice
- d) Lessons Learned

Tasks of the project manager

- Cost monitoring and budget adherence
- Adherence to deadlines and objectives
- Conflict management
- professional responsibility for the project team
- Management of general project activities
- Compliance with guidelines, procedures and work instructions

Tasks in non-agile implementation

- Communication within the team and with the client
- Preparation and execution of project meetings
- Creation and updating of the project plan